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# EECS 16A Touchscreen 2

**\*\*TA, UCS1, UCS1, UCS1\*\***

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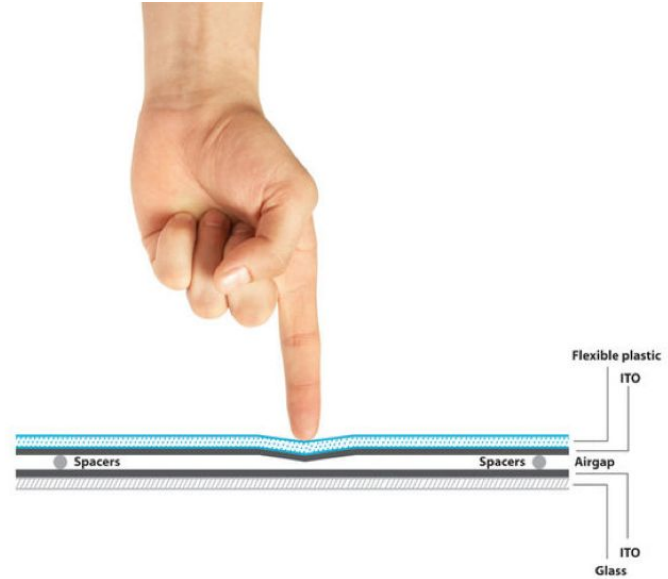
# Resistive Touchscreen

- Investigate a resistive touchscreen
  - Something that actually was used for a long time!
  - Top: Flexible resistive layer
  - Bottom: Our resistor circuit layer.
- Use voltage as a signal to determine position of touch
  - How?

# Resistive Touchscreen

- Physical touch results in physical contact between top and bottom layers
- Voltage dividers allow us to compute touch location

EX: Nokia N900, Nokia N97  
Mini, LG Optimus, LG GW620,  
Nintendo DS <sup>TM</sup>



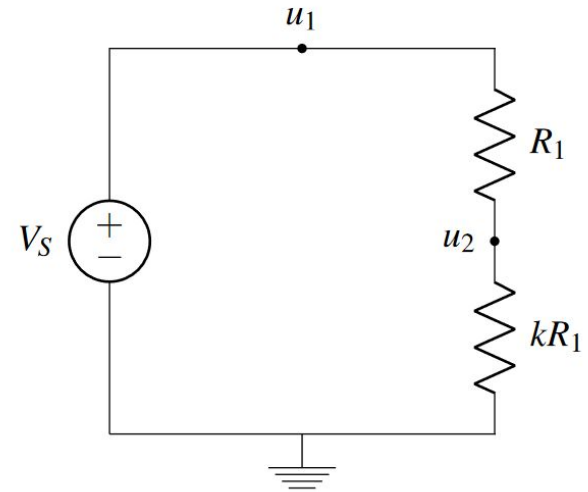
Resistive touchscreen

# Tools for Today:

- Power Supply (Always set a current limit of 0.1 A!)
- Multimeter - measuring device
- Soldering iron and PCBs!
- Voltage dividers
  - How we will detect location
- Falstad
  - Circuit simulation, has virtual Power Supplies and Multimeters

# Touchscreen Theory (Note 13/14)

- What's the voltage at the top?
- What's the voltage at the bottom?
- Voltage at  $u_2$ ?



# Touchscreen Theory (Note 13/14)

- What's the voltage at the top?

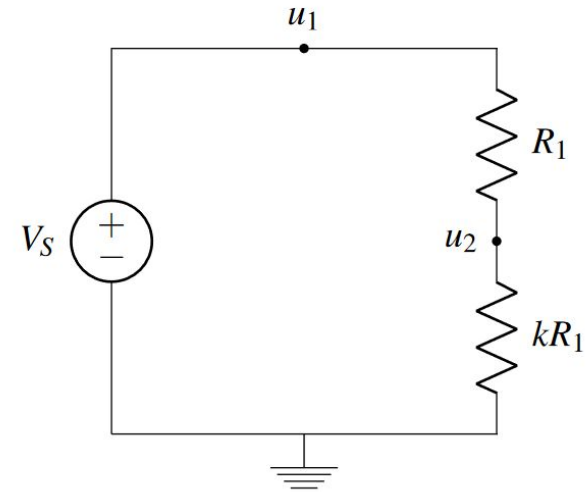
$V_S$

- What's the voltage at the bottom?

0

- Voltage at  $u_2$ ?

Voltage divider!



# Touchscreen Theory (Note 13/14)

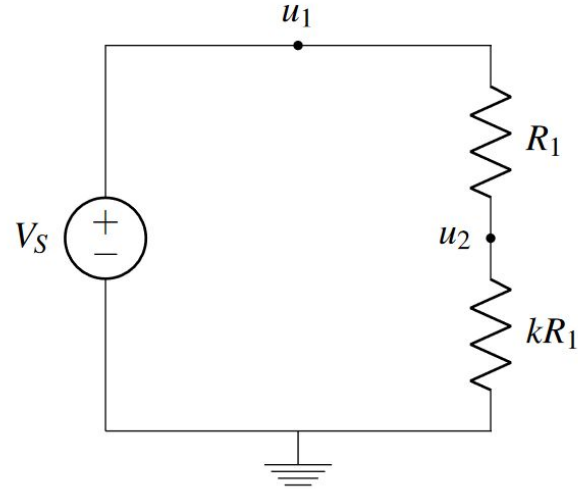
- Voltage divider:

$$u_2 = V_s * \frac{kR_1}{kR_1 + R_1}$$

$$u_2 = V_s * \frac{R_1(k)}{R_1(k+1)}$$

$$u_2 = V_s * \frac{k}{k+1}$$

*Independent of  
the value of R!*



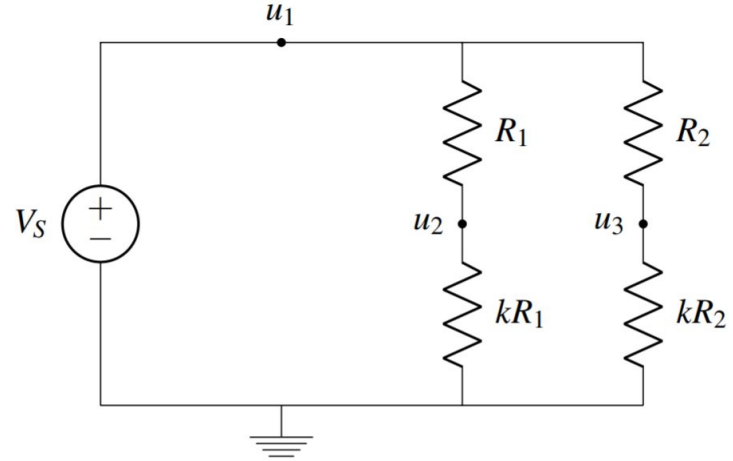
# Building it up

- What are the voltages at  $u_2$  and  $u_3$ ?

$$u_2 = V_s * \frac{k}{k + 1}$$

$$u_3 = V_s * \frac{k}{k + 1}$$

- What's the voltage difference?**



The  $R$ s cancel out! All that matters is the proportion between the top and bottom resistors.

In fact,  **$u_3$  and  $u_2$  are at the SAME VOLTAGE**

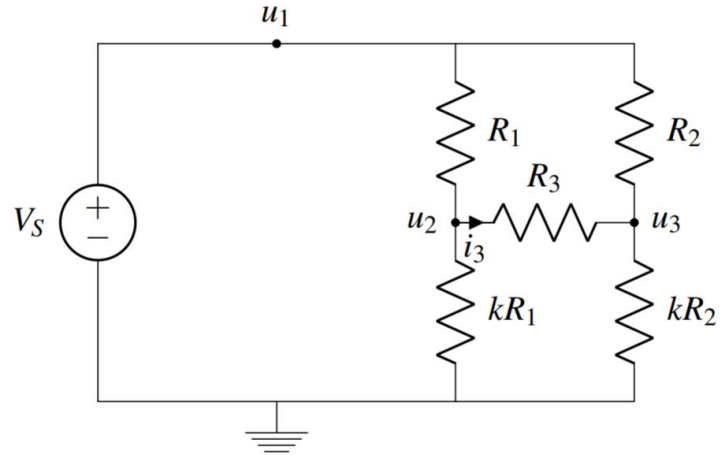


# Building it up

- We know that  $u_2 - u_3 = 0$
- **How much current goes through  $R_3$ ?**

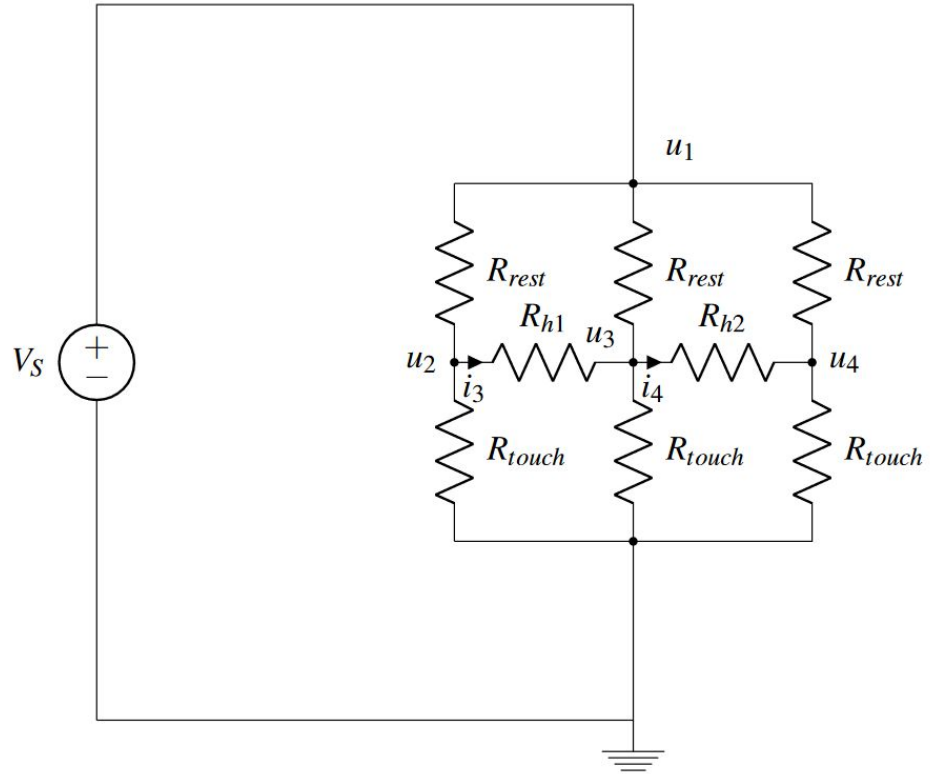
$$u_2 = V_s * \frac{k}{k+1}$$

$$u_3 = V_s * \frac{k}{k+1}$$



# Building it up

- Add one more resistor divider...
- We get our touchscreen!



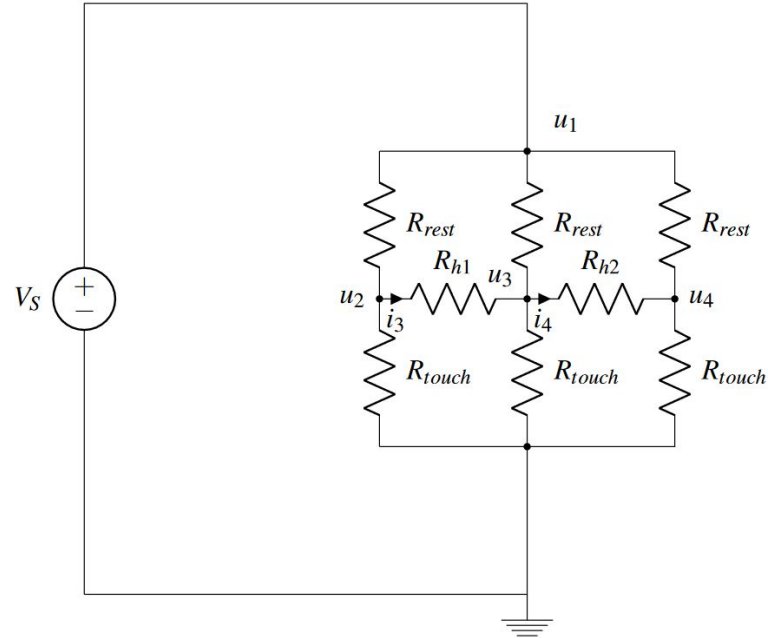
# Poll Time!

What is the voltage at  $u_4$ ?

- 0V
- Same as  $u_2$
- None of the above

How much current is flowing through  $R_{h2}$ ?

- 0A
- Non-zero current



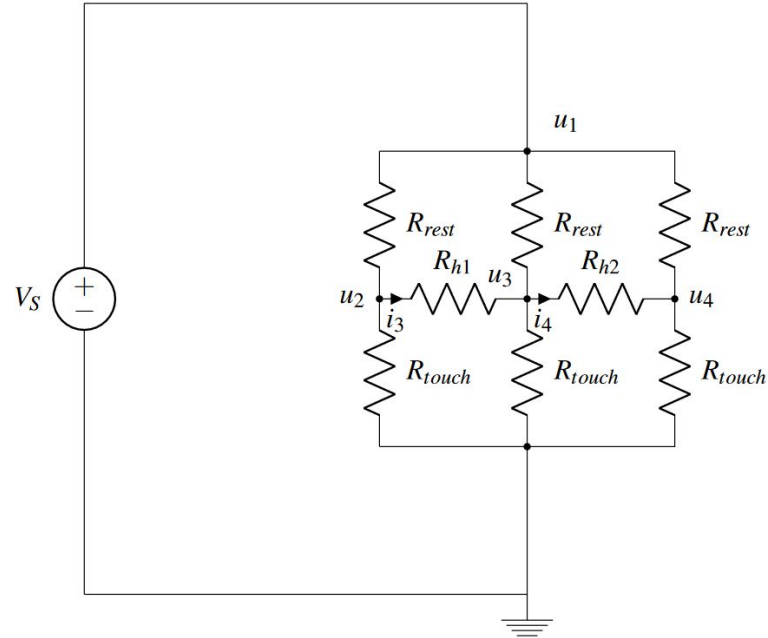
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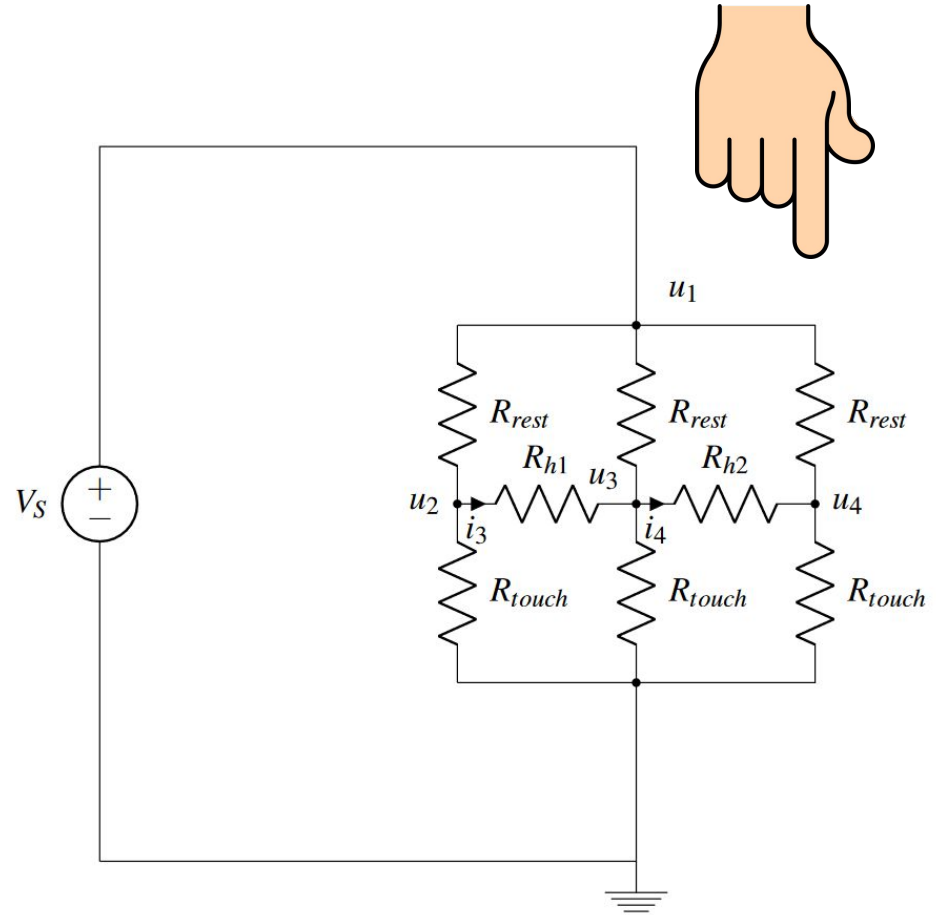
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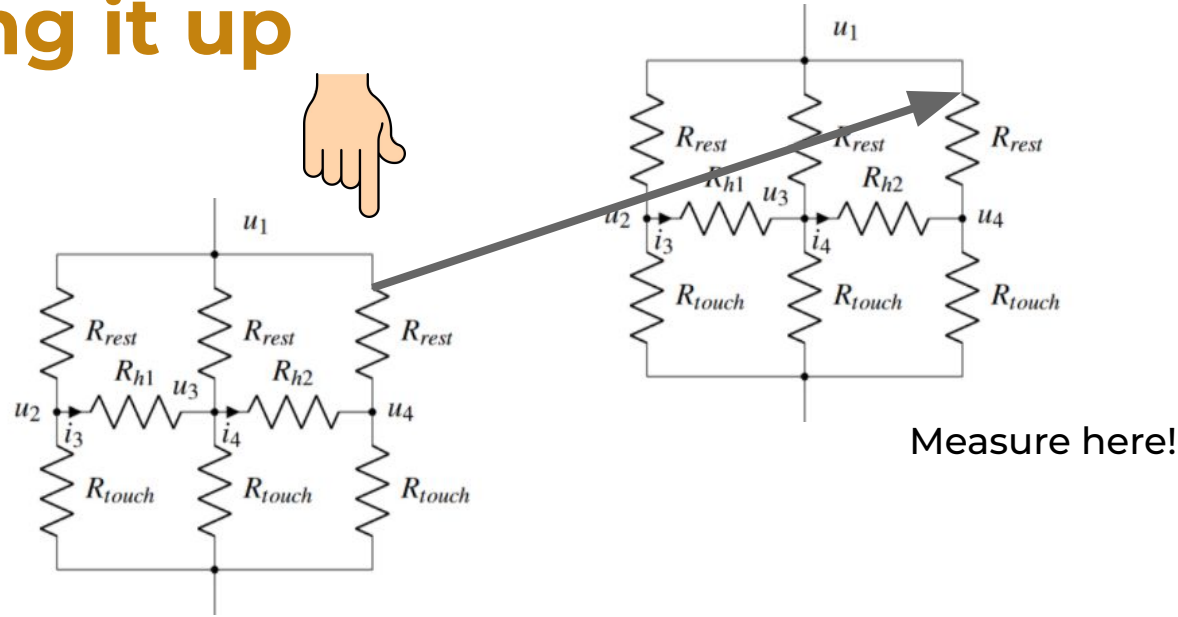


# Building it up

- But how do we measure the voltage?
- Our finger can press down on a point, but we need the voltage measurement!

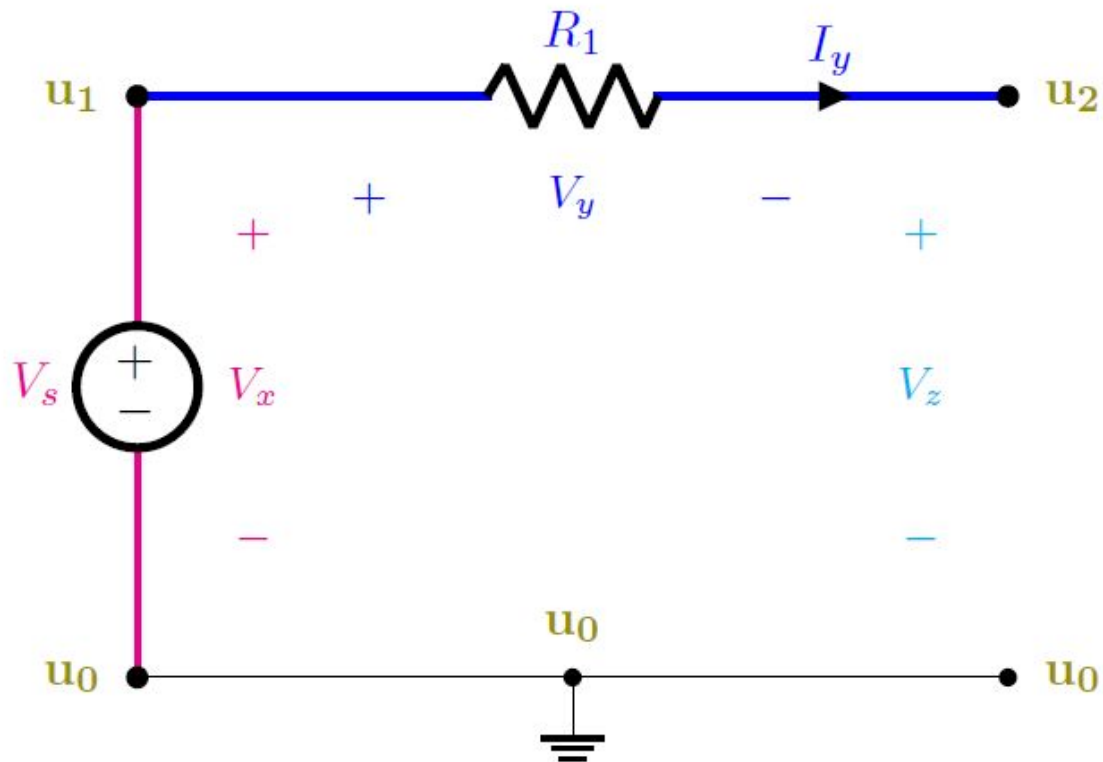


# Building it up



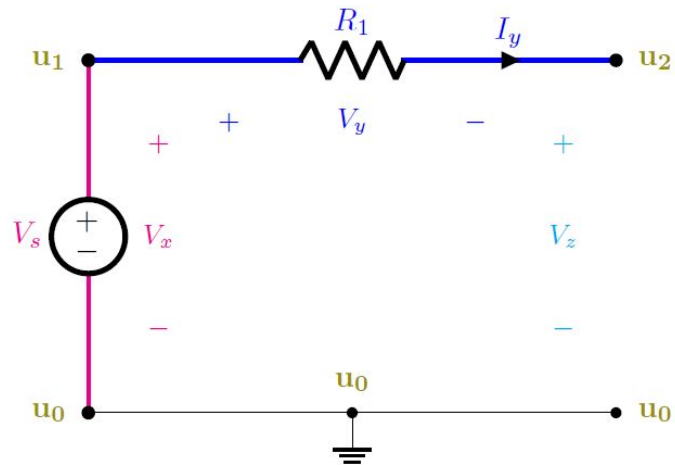
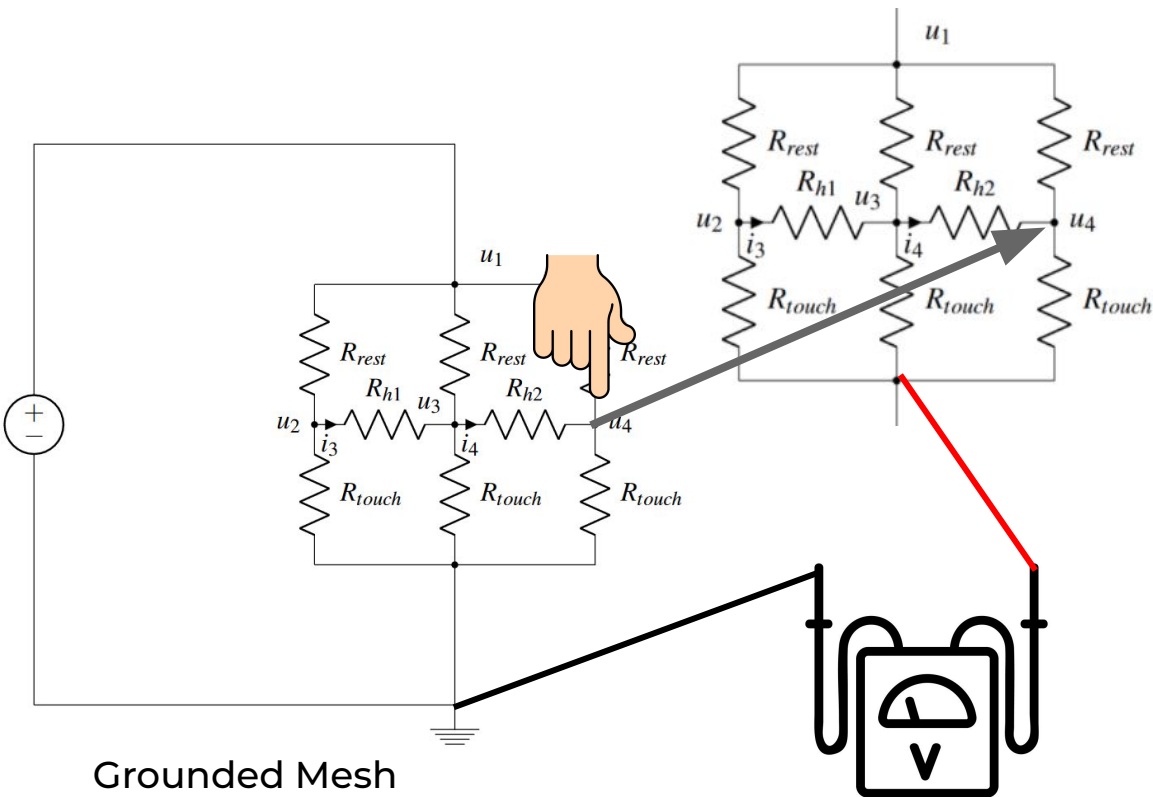
- We can add another (ungrounded) mesh!
- If we connect the meshes at the point we touch, we get the voltage all over the added (ungrounded) mesh!
- Why specifically a mesh? We'll see in a bit.

# Recall Touch 1



# Building it up

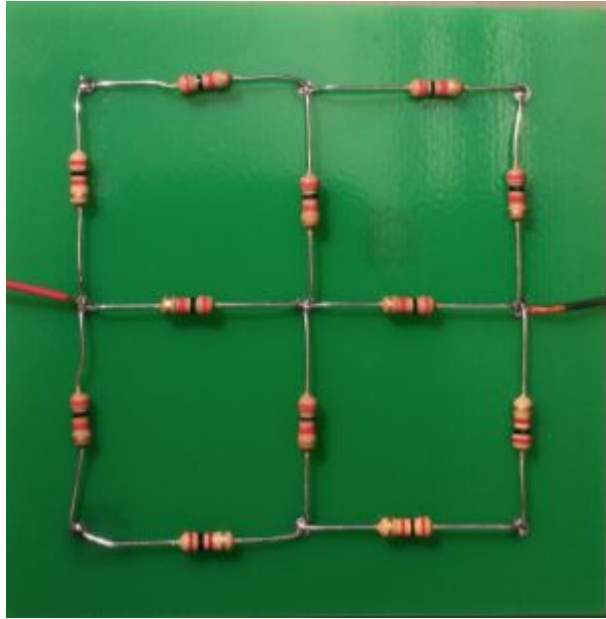
## Ungrounded Mesh





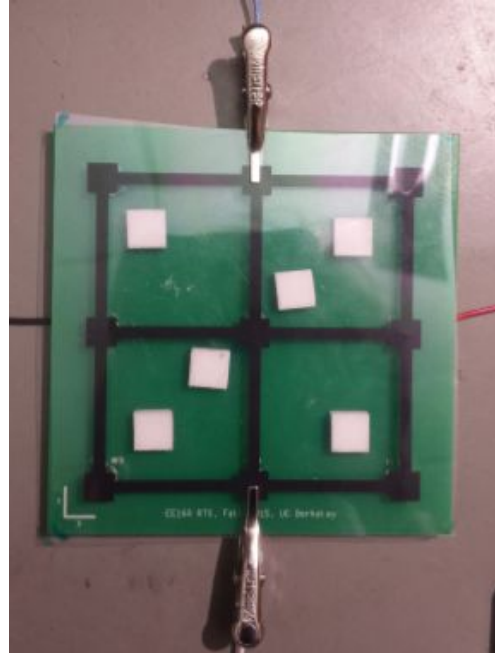
# Resistive Touchscreen - 2 Layers

Bottom Layer: Resistive Layer



# Resistive Touchscreen - 2 Layers

Top Layer: Flexible Resistive Layer

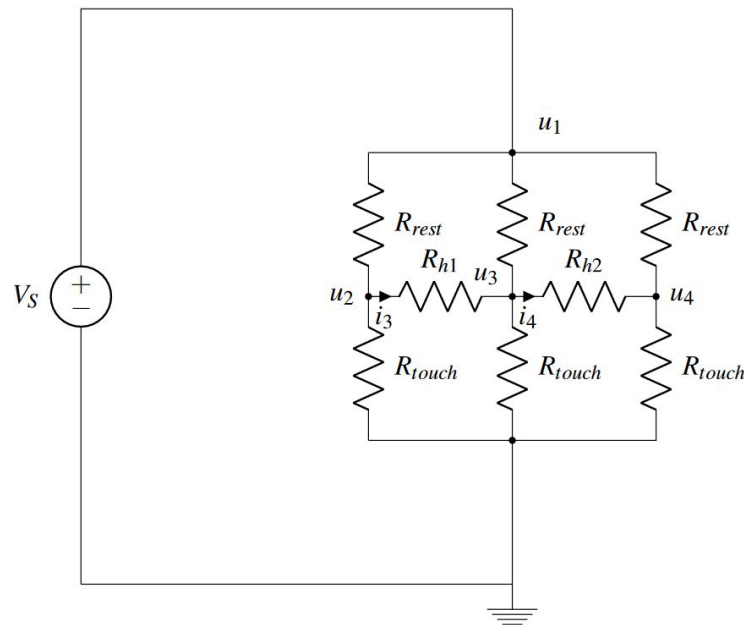


# What's the difference?

- Nothing
  - The ink is a bunch of resistors
    - The resistor values don't matter because we showed only the proportions matter for this circuit
  - Their circuit diagrams are the same
- One is flexible so we can actually move it to make contact
- We use two so that we can measure with one and apply voltage to the other without changing our circuit

# Computing a Location

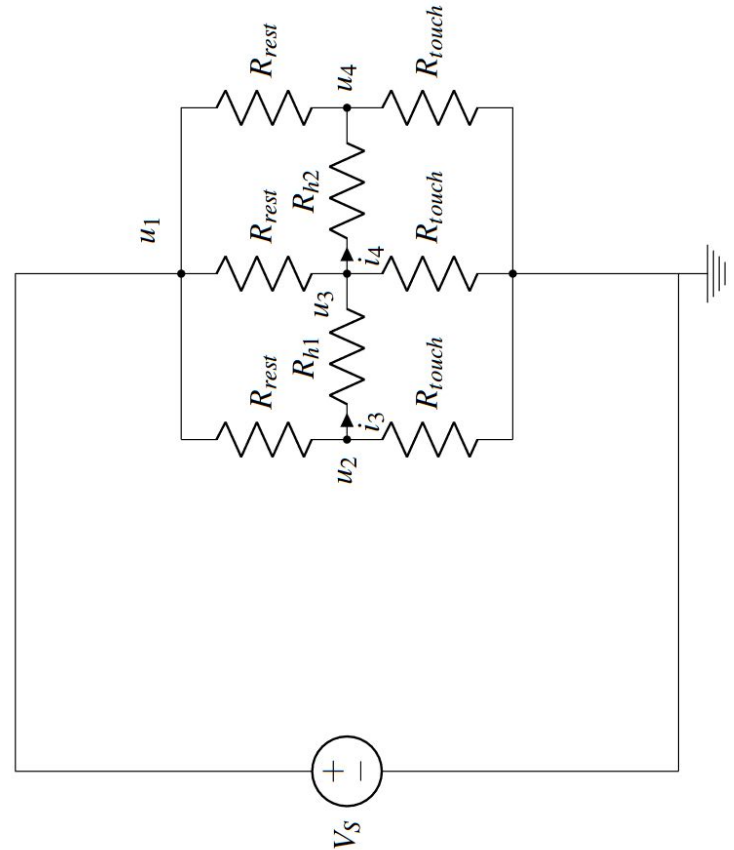
- Measure some voltages, compute location based on value
- **Can you find any two horizontal locations that would output the same voltage?**
- **What about vertical?**



# Computing a Location

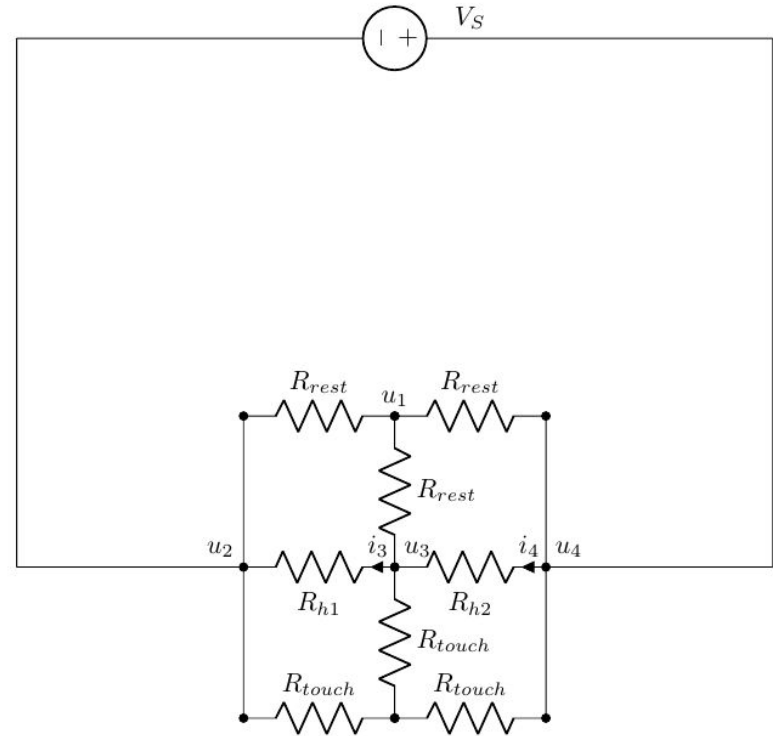
- We can only determine vertical position
- So, what about this other orientation?

What if we  
turned it  
sideways?

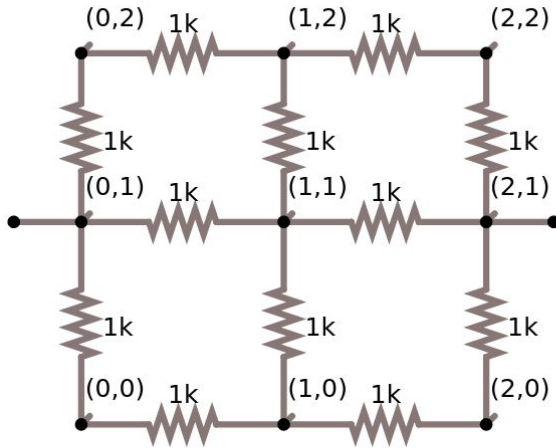


# Computing a Location

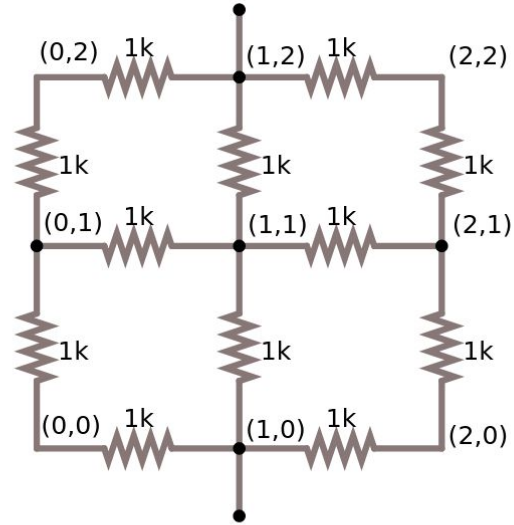
- Let's turn it sideways
  - Apply voltage so we power the horizontal direction
  - **Now, we can find vertical locations that would output the same voltage**
  - **But we cannot find horizontal locations that would output the same voltage**
- This lets us determine horizontal location



# Computing a location



Bottom mesh

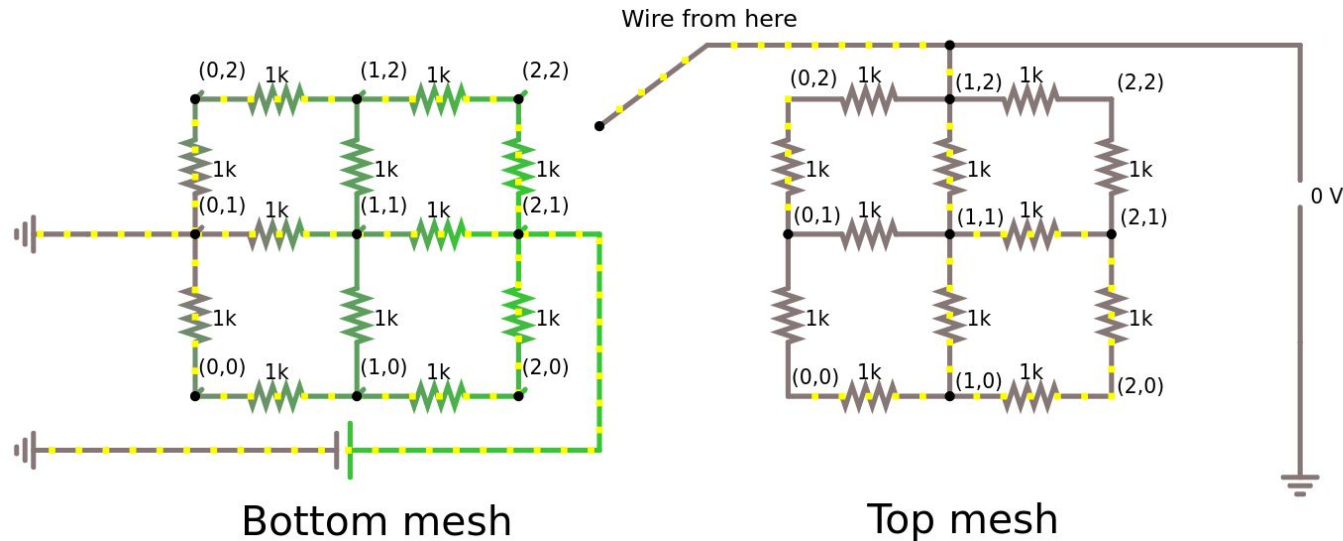


Top mesh

- Additional resistors? Combine as equivalent resistors in series!

# Computing a Location

- If we take two readings, one in each dimension can uniquely determine our location in 2D
- More on this in the lab notebook



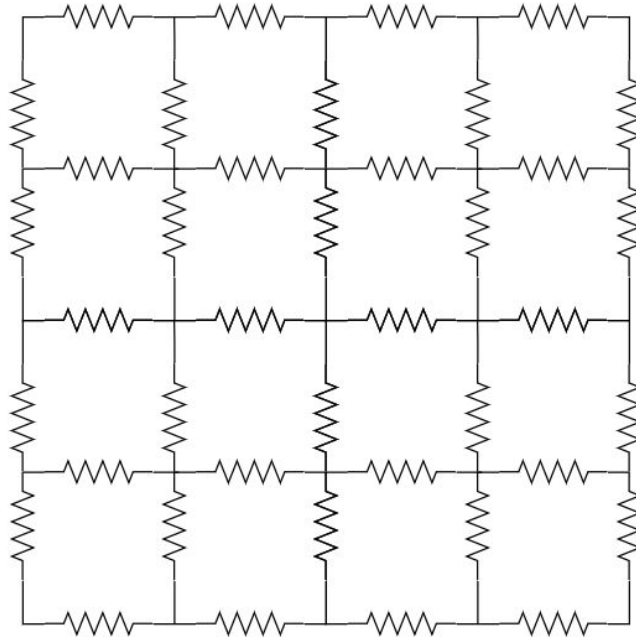


# Taking the Limit

- 9 touch points is kinda... meh
- **How do we get more?**

# Taking the Limit

- Add more resistors!

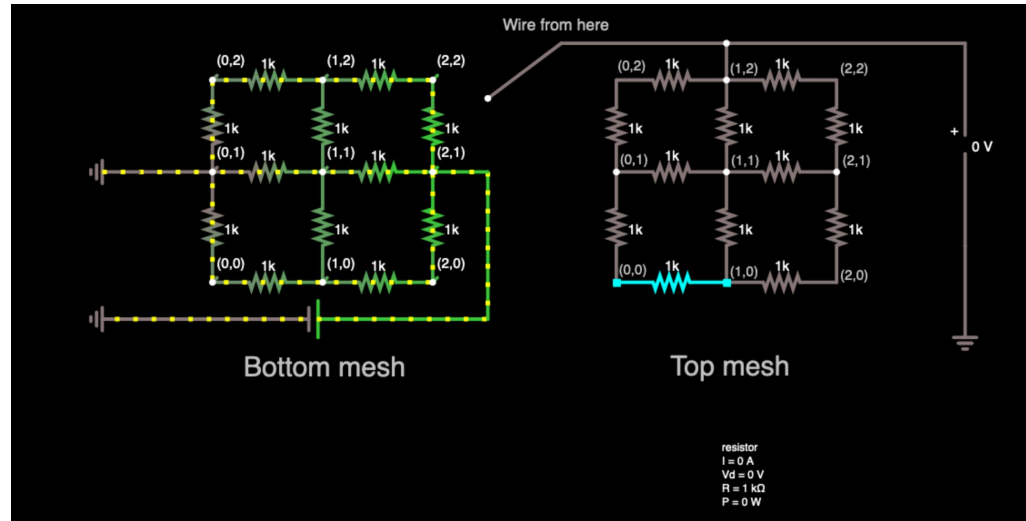


# Taking the Limit

- But what if I don't want to increase the size of the circuit?
  - Add more, but make the resistors smaller!
- What happens as the resistors approach infinitely small sizes?
  - Isn't that just a resistive sheet?
  - This is how all resistive touchscreens work
  - Review lecture [note 12](#), [note 13](#), [note 14](#)

# Simulating Touchscreens

- Falstad simulator ([Link](#))
  - Will be used in this lab to simulate resistive dividers in upper and bottom plates



# Building the Mesh

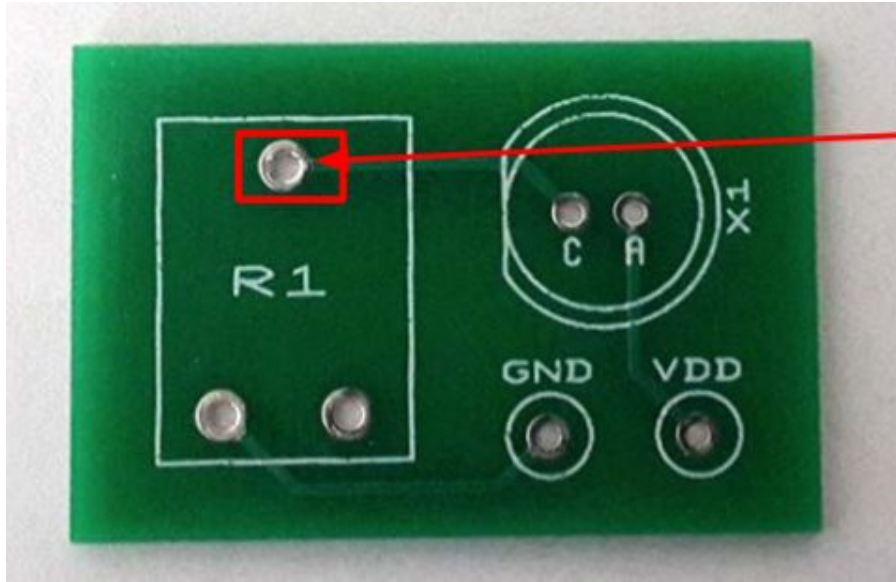
- Today's lab will have you build your own 3x3 resistive touchscreen almost from scratch!
- You'll need to learn how to solder in order to build your mesh
  - Be safe, let us know if you are unsure or if you get hurt
  - First aid at the TA desk

# Soldering



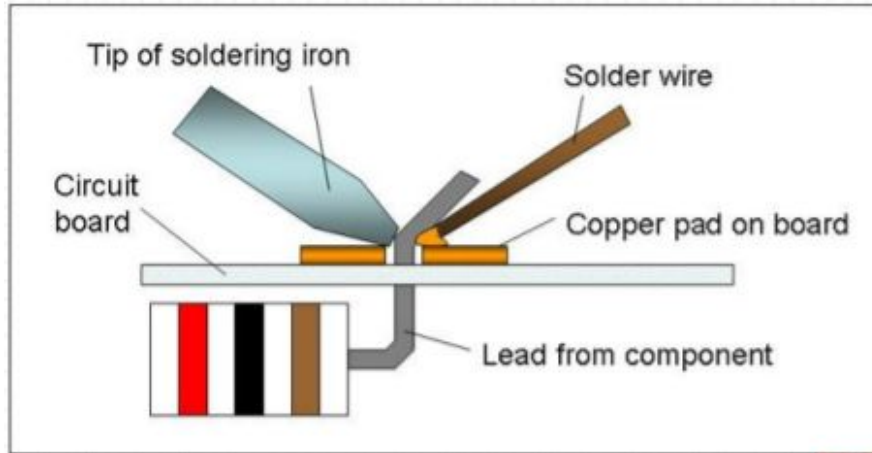
- Like hot glue for circuits
- Ensures there's a physical connection for all your components

# PCB (Printed Circuit Board)

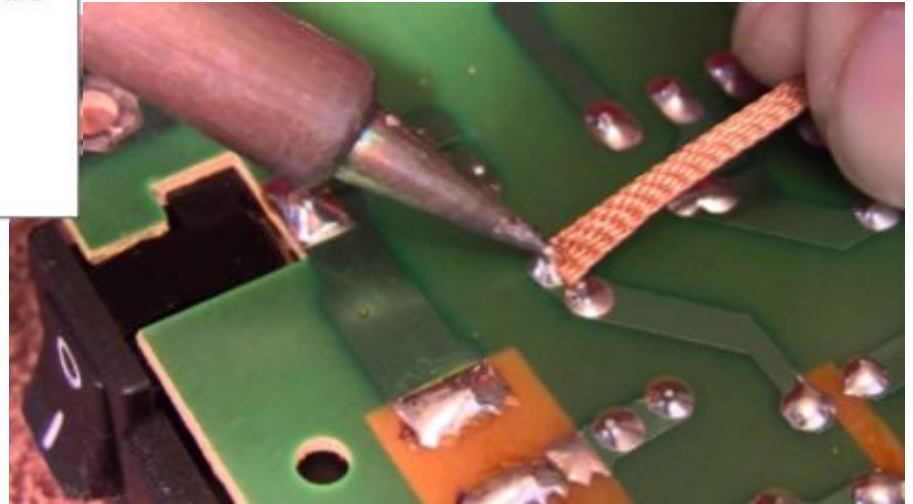


Pad (Copper plate)

# Soldering (Cont.)

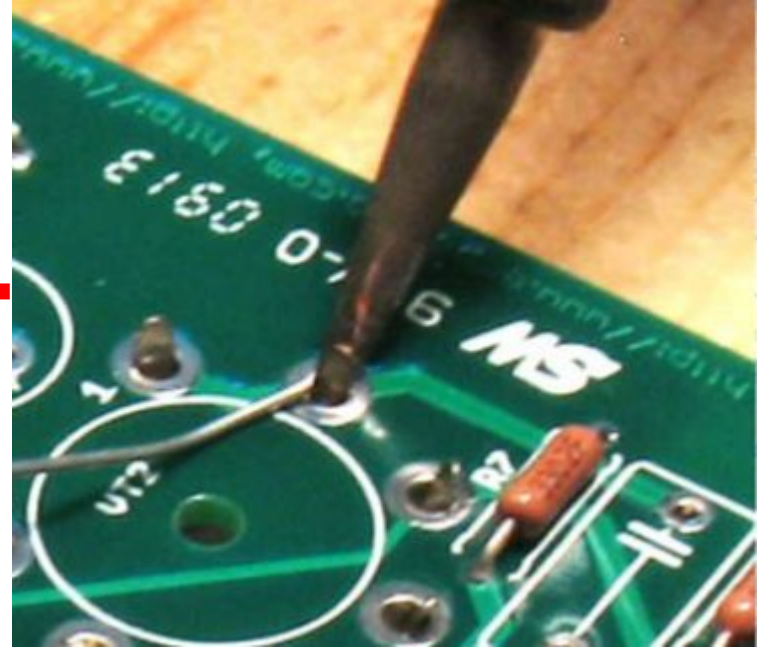
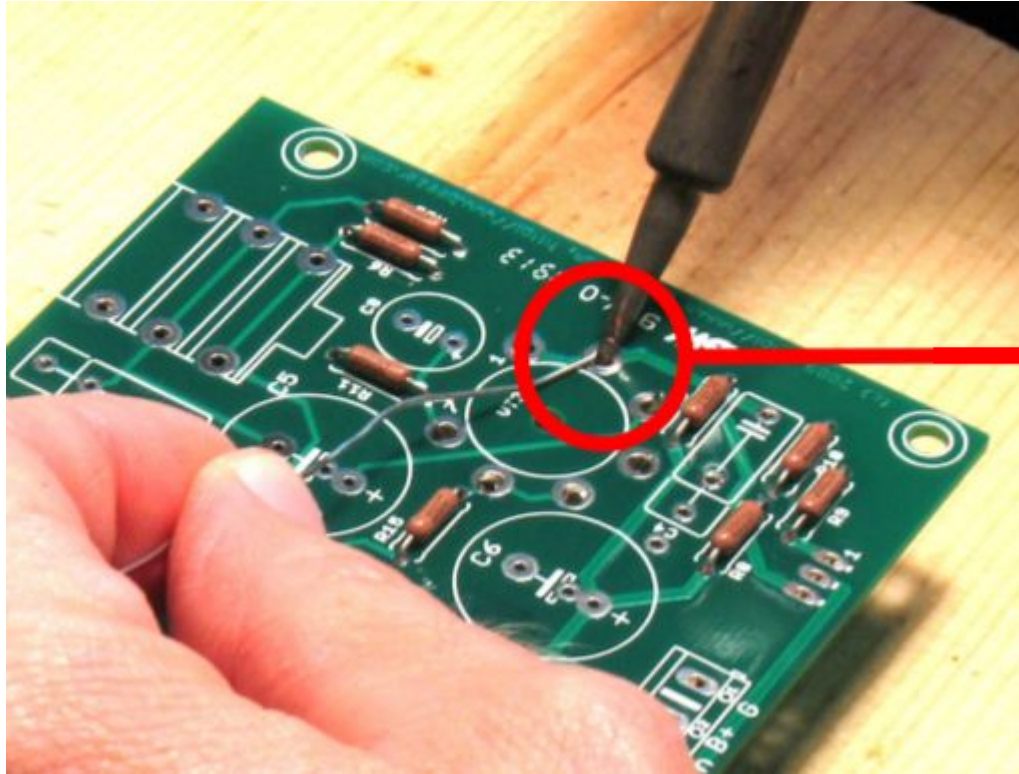


Only solder metal to metal!

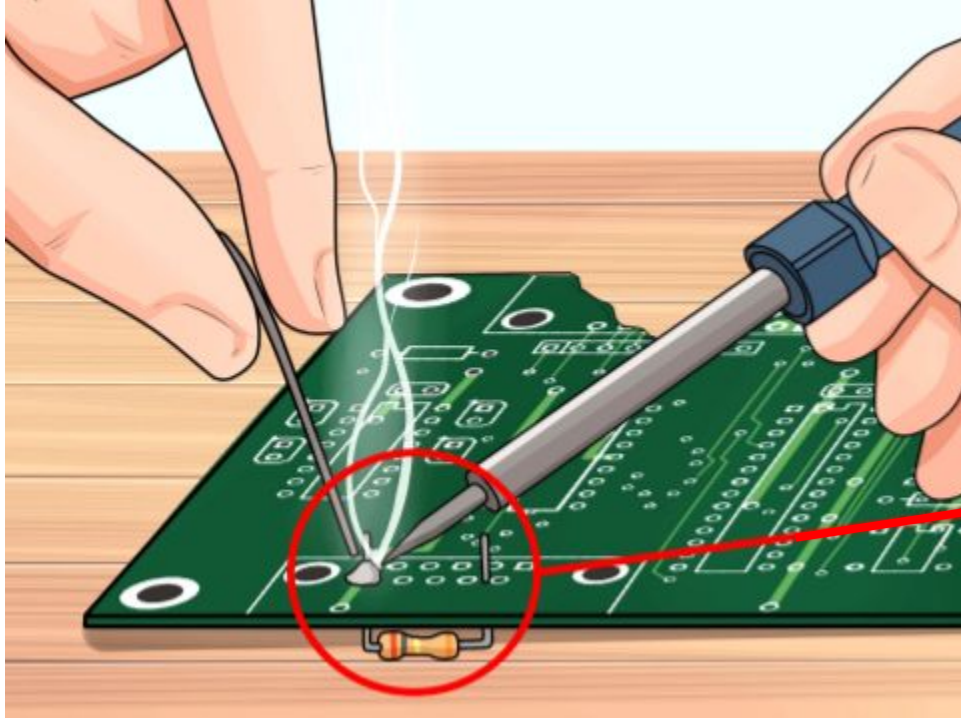




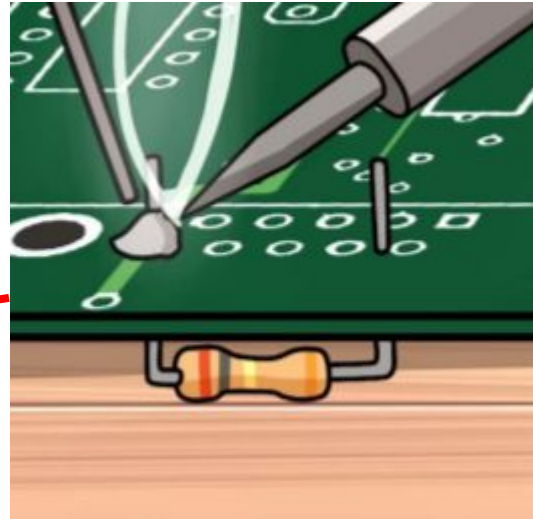
# Soldering (Cont.)



# Soldering (Cont.)



Component should be flush to the board...why?



# Soldering DON'TS

- **Never** eat in lab, but especially not today
- Don't solder jumper wires!
  - Use breadboarding wires from TA desk
- Don't solder plastic and other non-metals
  - Strip your wires! Wire strippers are at your station
- Don't use too much solder
  - But also not too little

# Soldering DO'S

- Wet the sponge and clean the tip when you start
- Use the clamp: components should be stable before you start soldering
  - Push components **ALL THE WAY IN**
- **Safety first**; don't burn yourself
  - If you do, let us know immediately!
- Ask if you're unsure about what you're doing
- Clean up after yourself, turn off all equipment

# Soldering Grip



# Soldering Intro Quiz Check

- Before you're allowed to begin soldering, you and your lab partner must show a TA/lab assistant your quiz score
  - One quiz per pair
  - Both students must be able to explain any incorrect answers

# Pac-Man!

- Thanks to the efforts of your wonderful lab development staff, you'll be able to play pac-man using your touchscreen at the end of the lab!
- **Please don't close the pac-man window – doing so will cause your kernel to crash.** Instead, just lose the game and it'll close automatically.

# Pointers

- Watch soldering videos for guidance
- Complete and get checked-off for the **soldering quiz** before soldering
- Water squirters, precision cutters, solder rolls **STAY** on the TA Desk
- **Don't solder jumper wires**
- **Don't forget to solder 2 standard wires to the board**
- Make sure components are all flat on the PCBs
- Cut the soldered joints **VERY SHORT**
- Turn off your soldering iron when done soldering

Checkoff Form:



Feedback Form:





# Feedback

For lab training, submit your feedback here:

<https://tinyurl.com/fb-training-fa23>

Touch 2:

<https://tinyurl.com/fa23-touch2>

Reminder:

- TA Post-Lab Feedback:

<https://tinyurl.com/fb-ta-fa23>

